

Plant Model: Full Nonlinear Formulation and its Linear Approximation

Synthetic Fuel Production — `sfp`

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Section 1 is the model integrated by the simulator and differentiated by the inner NMPC — one set of equations, used by both, so there is no second copy to drift. Section 2 is the reduced linear model the outer dispatch layer optimises. Parameter provenance is tagged in `docs/ASSUMPTIONS.md`; values quoted here are the defaults in `sfp/models/params/*.yaml`. Two thirds of them are **assumed**.

1 Full nonlinear model

1.1 Conventions

The plant state is $x \in \mathbb{R}^{16}$, the control $u \in \mathbb{R}^{11}$, and weather w carries plane-of-array irradiance G (W/m^2), air temperature T_a ($^\circ\text{C}$), wind speed v (m/s), relative humidity and pressure. Each machine takes a *(setpoint, enable)* pair; the battery takes two powers and the array a curtailment fraction.

Sign convention: positive electrical power is power *consumed* from the DC bus. The array is therefore a negative load and battery discharge is negative.

Three smoothing primitives appear throughout, all from `sfp/models/mathx.py`. They exist because the inner layer differentiates these expressions and a hard switch has no usable gradient:

$$\text{sstep}(z; \varepsilon) = \frac{1}{2}(1 + \tanh(z/\varepsilon)), \quad \text{sclip}(z; a, b, \varepsilon) \approx \min(\max(z, a), b). \quad (1)$$

Every machine is gated by $g(u) = \text{sstep}(u_{\text{enable}} - \frac{1}{2}; 0.05)$, which is 2.06×10^{-9} when off and 1 when on. Integration is fixed-step RK4 with a zero-order hold on u and w .

1.2 PV array

No states. Input: curtailment fraction $\gamma \in [0, 1]$.

Faiman (2008) cell temperature, linear temperature derating about STC, constant inverter efficiency and a hard clip at the AC rating:

$$T_{\text{cell}} = T_a + \frac{G}{u_0 + u_1 v}, \quad f_{\text{derate}} = (1 - s)(1 - \ell)(1 - d)^y, \quad (2)$$

$$P_{\text{dc}} = P_{\text{dc}}^{\text{rated}} \frac{G}{G_{\text{ref}}} (1 + \gamma_P (T_{\text{cell}} - T_{\text{ref}})) f_{\text{derate}}, \quad P_{\text{ac}} = \min(\eta_{\text{inv}} P_{\text{dc}}, P_{\text{ac}}^{\text{rated}}), \quad (3)$$

with $P_{\text{ac}}^{\text{rated}} = P_{\text{dc}}^{\text{rated}} / \text{DC:AC}$ and delivered power $(1 - \gamma)P_{\text{ac}}$. Clipping loss and deliberate curtailment are reported separately — the brief asks for curtailment, and off grid it is simply PV that cannot be absorbed.

γ_P	power temperature coefficient	-0.0035 K^{-1}
$T_{\text{ref}}, G_{\text{ref}}$	STC reference	$25 \text{ }^\circ\text{C}, 1000 \text{ W}/\text{m}^2$
u_0, u_1	Faiman coefficients	$25 \text{ W}/(\text{m}^2\text{K}), 6.84 \text{ Ws}/(\text{m}^3\text{K})$
$\eta_{\text{inv}}, \text{DC:AC}$	inverter	$0.975, 1.25$
s, ℓ, d	soiling, wiring, annual degradation	$0.03, 0.02, 0.005 \text{ yr}^{-1}$

1.3 Battery

States: state of charge σ , fractional capacity fade ϕ , cumulative equivalent full cycles n . Inputs: charge and discharge power, both non-negative.

$$E_{\text{use}} = E_{\text{nom}} \max(1 - \phi, 10^{-3}), \quad \dot{\sigma} = \frac{\eta_c P_{\text{ch}} - P_{\text{dis}}/\eta_d}{E_{\text{use}}} - \frac{\kappa_{\text{sd}}}{86400} \sigma, \quad (4)$$

$$\dot{n} = \frac{P_{\text{ch}} + P_{\text{dis}}}{2E_{\text{nom}}}, \quad \dot{\phi} = \frac{\phi_{\text{eol}}}{N_{\text{cyc}}} S(\sigma) \dot{n} + \frac{\phi_{\text{cal}}}{T_{\text{yr}}}. \quad (5)$$

Charge and discharge are *separate non-negative* variables rather than one signed power. Nothing forbids both being positive at once; what prevents it is that the wear term makes it strictly worse than either alone, which is how this formulation avoids an explicit complementarity constraint.

The depth-of-discharge stress factor S encodes $N(\text{DoD}) = N_{\text{ref}}(\text{DoD}_{\text{ref}}/\text{DoD})^k$:

$$S(\sigma) = C \left[\underbrace{\max(2|\sigma - \bar{\sigma}|/\text{DoD}_{\text{ref}}, f_{\text{min}})}_{\text{depth}} \right]^{k-1}, \quad \bar{\sigma} = \frac{1}{2}(\sigma_{\text{max}} + \sigma_{\text{min}}), \quad (6)$$

with C chosen so a cycle at the rated depth costs exactly one rated cycle. **This is a local proxy for a path-dependent quantity:** depth of discharge is properly a property of a *cycle*, recovered by rainflow counting, and a differential model cannot see a cycle. Without it, throughput is all that matters and a large pack swinging gently is charged the same per kWh as a small one cycling to its limits — which flatters small packs precisely in a sizing sweep.

η_c, η_d	one-way efficiencies	0.96, 0.96
$\sigma_{\text{min}}, \sigma_{\text{max}}$	SoC window	0.10, 0.95
κ_{sd}	self-discharge	0.002 d ⁻¹
$N_{\text{cyc}}, \text{DoD}_{\text{ref}}$	rated cycle life	6000 at 0.80
k, f_{min}	DoD exponent, stress floor	3.0, 0.25
$\phi_{\text{eol}}, \phi_{\text{cal}}$	end-of-life fade, calendar fade	0.20, 0.015 yr ⁻¹

1.4 Air contactor

No states — the sorbent it loads lives in §1.5. Input: airflow fraction φ and enable.

Capture and fan power scale oppositely with airflow, which is the whole point of the subsystem:

$$V = \varphi g(u) V_{\text{max}}, \quad \rho = \frac{p}{RT_a}, \quad \text{NTU} = \text{NTU}_d \frac{V_d}{\max(V, 10^{-3})}, \quad (7)$$

$$\eta_{\text{cap}} = 1 - e^{-\text{NTU}}, \quad f_T = \exp\left[-\frac{E_a}{R}\left(\frac{1}{T_a} - \frac{1}{T_{\text{ref}}}\right)\right], \quad f_{\text{RH}} = \frac{\text{rh}/(\text{rh} + k_h)}{\text{rh}_{\text{ref}}/(\text{rh}_{\text{ref}} + k_h)}, \quad (8)$$

$$r_{\text{carb}} = V \rho y_{\text{CO}_2} \eta_{\text{cap}} f_T f_{\text{RH}} f_{\text{load}}, \quad P_{\text{fan}} = P_{\text{fan}}^{\text{rated}} \varphi^3 + P_{\text{idle}} g(u). \quad (9)$$

$\text{NTU}_d = -\ln(1 - \eta_d)$; residence time falls as $1/V$ in a fixed bed, so $\text{NTU} \propto 1/V$. The load taper $f_{\text{load}} = \text{sclip}((1 - L)/(1 - L_{\text{taper}}))$ stops the fan pushing CO₂ into a saturated bed.

Diminishing returns are severe. At reference sizing, 20°C and 60% RH: 60% flow captures 0.233 mol/s for 4.7 kW (128 kWh/tCO₂); 100% flow captures 0.279 mol/s for 20.4 kW (462 kWh/tCO₂). Twenty percent more CO₂ for four and a third times the fan power — a power-follow controller that dumps surplus solar into the fans burns most of it for nothing.

V_{max}, V_d	max and design airflow	40, 24 m ³ /s
η_d	design single-pass capture	0.65
$P_{\text{fan}}^{\text{rated}}, P_{\text{idle}}$	fan power	20 kW, 0.4 kW
y_{CO_2}	ambient mole fraction	420 ppm (measured)
$E_a, k_h, L_{\text{taper}}$	activation, RH half-sat., taper	20 kJ/mol, 0.3, 0.9

1.5 Sorbent inventory

States: n_{CaO} , n_{CaCO_3} , mean cycle number $\bar{N}$. No inputs — it is driven by the contactor and calciner rates.

$$\dot{n}_{\text{CaO}} = r_{\text{calc}} - r_{\text{carb}} + \dot{m} - \pi r_{\text{calc}}, \quad \dot{n}_{\text{CaCO}_3} = r_{\text{carb}} - r_{\text{calc}}, \quad \dot{\bar{N}} = \frac{r_{\text{calc}} - \dot{m}\bar{N}}{n_{\text{tot}}}. \quad (10)$$

One full pass of the inventory through the kiln advances $\bar{N}$ by exactly one; make-up dilutes the mean age. Grasa–Abanades deactivation sets the capacity:

$$X(\bar{N}) = \frac{1}{k\bar{N} + (1 - X_r)^{-1}} + X_r, \quad X_r = 0.075, \quad k = 0.52. \quad (11)$$

$\bar{N}$ is kept *continuous* rather than integer specifically so it stays differentiable inside the NLP. Because $dX/d\bar{N} < 0$, every calcination permanently shrinks the capacity of the *whole* inventory — the marginal cost the controller pays per mole calcined, and the reason it is reluctant to cycle the loop harder than the methane is worth.

$n_{\text{tot}} = 50\,000$ mol; make-up and purge are both zero by default, so deactivation is irreversible in the runs reported.

1.6 Calciner

State: kiln temperature T . Inputs: heater fraction and enable.

Three independent limits multiply — thermodynamics, kinetics, feedstock:

$$p_{\text{eq}}(T) = A_B e^{-B_B/T} \quad (\text{Baker 1962}), \quad f_{\text{drive}} = \text{sclip}\left(1 - \frac{p_{\text{op}}}{p_{\text{eq}}}\right), \quad (12)$$

$$f_{\text{kin}} = A_k e^{-E_a/(RT)}, \quad f_{\text{feed}} = \text{sclip}\left(\frac{n_{\text{CaCO}_3}}{n_{\text{ref}}}\right), \quad (13)$$

$$r_{\text{calc}} = r_{\text{max}} f_{\text{drive}} \text{sclip}(f_{\text{kin}}) f_{\text{feed}} g(u). \quad (14)$$

Energy balance, with all four terms in watts:

$$C \dot{T} = \underbrace{f_h P_h^{\text{rated}} \eta_h}_{\text{heater}} - \underbrace{(UA + c_v v)(T - T_a)}_{\text{loss}} - \underbrace{r_{\text{calc}} \Delta H_{\text{calc}}}_{\text{reaction}} - \underbrace{r_{\text{calc}} M_{\text{CaCO}_3} c_p \max(T - T_{\text{feed}}, 0)}_{\text{sensible}}. \quad (15)$$

The heater fraction carries an over-temperature interlock. **The kiln’s thermal inertia is the plant’s commitment problem:** $C = 3 \times 10^6$ J/K against a 150 kW heater means **5.6 hours from ambient to 1200 K**, so a kiln worth lighting is a kiln worth running for a while. That single fact is why the outer layer keeps a kiln temperature state at all.

$C, P_h^{\text{rated}}, \eta_h$	thermal capacity, heater	3.0×10^6 J/K, 150 kW, 0.95
UA, c_v	loss coefficient, wind term	17 W/K, 0.6 W/(K m/s)
A_B, B_B	Baker equilibrium	4.137×10^7 atm, 20474 K
p_{op}	kiln operating pressure	0.3 atm
A_k, E_a	Arrhenius	2.207×10^5 s ⁻¹ , 120 kJ/mol
$r_{\text{max}}, T_{\text{max}}$	feeder throughput, limit	0.45 mol/s, 1273 K
ΔH_{calc}	endothermic	+178 kJ/mol

1.7 PEM electrolyser

States: stack temperature T , voltage degradation δV . Input: current fraction and enable. $i = \varphi g(u) i_{\text{max}}$.

$$V_{\text{rev}}(T) = V_{\text{rev}}^0 + \alpha_T(T - 298.15), \quad \eta_{\text{act}} = \frac{RT}{\alpha_c F} \text{asinh}\left(\frac{i}{2i_0}\right), \quad (16)$$

$$\text{ASR}(T) = \text{ASR}_0(1 + \beta(T - T_{\text{set}})), \quad V_{\text{cell}} = V_{\text{rev}} + \eta_{\text{act}} + i \text{ASR} + \delta V. \quad (17)$$

The activation term is Butler–Volmer in asinh form, which is exact and—unlike the Tafel approximation—smooth down to zero current.

$$r_{\text{H}_2} = \frac{\eta_F N_c i A}{2F}, \quad P_{\text{stack}} = N_c V_{\text{cell}} i A, \quad Q_{\text{gen}} = N_c i A (V_{\text{cell}} - V_{\text{tn}}), \quad (18)$$

$$P_{\text{total}} = P_{\text{stack}} + P_{\text{aux}} g(u) + \kappa Q_{\text{gen}}, \quad C\dot{T} = Q_{\text{gen}} - Q_{\text{cool}} - UA(T - T_a), \quad \delta\dot{V} = \delta_r \varphi. \quad (19)$$

Heat is written as the voltage difference $V_{\text{cell}} - V_{\text{tn}}$ rather than (electrical in – chemical out), to avoid differencing two large, nearly equal numbers.

Efficiency peaks at part load and then falls, because η_{act} grows like $\log i$ while the ohmic term grows like i . “Run flat out when sunny” is therefore provably suboptimal — one of the few places where the economics and the physics point the same way.

N_c, A, i_{max}	cells, area, max current density	105, 1000 cm ² , 1.5 A/cm ²
i_{min}	minimum load (gas crossover)	0.10 i_{max}
$V_{\text{rev}}^0, V_{\text{tn}}, \alpha_T$	reversible, thermoneutral	1.229 V, 1.481 V, –0.9 mV/K
$i_0, \alpha_c, \text{ASR}_0, \beta$	kinetics, resistance	10 ^{–4} A/cm ² , 0.5, 0.15 Ωcm ² , –0.005 K ^{–1}
$\eta_F, P_{\text{aux}}, \kappa$	Faraday eff., auxiliaries, parasitic	0.98, 15 kW, 0.02
C, UA, T_{max}	thermal	2.5 × 10 ⁵ J/K, 150 W/K, 353.15 K

1.8 Gas buffers and water balance

Hydrogen and CO₂ tanks, no inputs:

$$\dot{n}_{\text{H}_2} = a_{\text{H}_2} r_{\text{ely}} - 4r_{\text{sab}}, \quad \dot{n}_{\text{CO}_2} = a_{\text{CO}_2} r_{\text{calc}} - r_{\text{sab}}, \quad (20)$$

where $a_{\bullet} \in [0, 1]$ are smooth acceptance gates that close as a tank fills. Production the tanks cannot accept is *vented and recorded*, not silently dropped from the mass balance: a power-follow controller can vent nearly half the hydrogen it makes, having paid ~56 kWh/kg for it, purely because it never looked at the tank level.

Water, with condensate recycle $\phi_r = 0.95$:

$$\dot{m}_w = \dot{m}_{\text{makeup}} + 2\phi_r M_{\text{H}_2\text{O}} r_{\text{sab}} - M_{\text{H}_2\text{O}} r_{\text{ely}}. \quad (21)$$

Net consumption is ≈2.25 kg H₂O per kg CH₄ after recycle — a real constraint at an arid site. Capacities: 30 000 mol H₂, 6000 mol CO₂, 5000 kg water; CO₂ compression costs 17.5 kJ/mol.

1.9 Sabatier reactor

States: temperature T , catalyst activity a , cumulative methane m . Input: feed fraction φ and enable.

$$X_{\text{eq}}(T) = \frac{1}{1 + \exp((T - T_{50})/w)}, \quad \text{Da} = \text{Da}_{\text{ref}} e^{-\frac{E_a}{R}(\frac{1}{T} - \frac{1}{T_{\text{ref}}})} \frac{a}{\max(\varphi, 0.02)}, \quad (22)$$

$$X = X_{\text{eq}}(1 - e^{-\text{Da}}), \quad r_{\text{sab}} = \varphi F_{\text{max}} X \Lambda(w). \quad (23)$$

X_{eq} *falls* with temperature because the reaction is exothermic; that is the term that stops the controller running the reactor as hot as the catalyst allows. Residence time rises as the feed is turned down, so Da rises — conversion improves at part load.

The availability factor is the hard stoichiometric coupling:

$$\Lambda = \min\left(\text{sclip}\left(\frac{n_{\text{CO}_2}}{50}\right), \text{sclip}\left(\frac{n_{\text{H}_2}}{200}\right)\right) \cdot \text{sstep}(m_w - 1). \quad (24)$$

Whichever of CO₂ or H₂ runs short throttles the reactor, no matter how much of the other is banked. This is what forces the two upstream chains to be co-scheduled rather than optimised separately, and it is the single most important coupling in the plant.

$$C\dot{T} = r_{\text{sab}}(-\Delta H) + Q_{\text{pre}} - Q_{\text{cool}} - UA(T - T_a) - Q_{\text{sens}}, \quad (25)$$

$$\dot{a} = -k_d \exp\left(\frac{T - T_{\text{ref}}^d}{T_{\text{scale}}}\right)a, \quad \dot{m} = r_{\text{sab}}M_{\text{CH}_4}. \quad (26)$$

Deactivation is exponential in temperature, so thermal excursions are permanently expensive.

F_{max}	max CO ₂ feed	0.12 mol/s
T_{50}, w	equilibrium logistic	810 K, 70 K
$\text{Da}_{\text{ref}}, E_a$	kinetics at 573 K	6.0, 80 kJ/mol
$T_{\text{ign}}, T_{\text{target}}, T_{\text{max}}$	light-off, target, sintering	523, 573, 823 K
k_d, T_{scale}	deactivation	10^{-8} s^{-1} , 50 K
ΔH	exothermic	-165 kJ/mol

1.10 DC bus

The subsystems are coupled only through the power balance, evaluated in registration order so that each publishes what the next needs:

$$\underbrace{P_{\text{fan}} + P_{\text{kiln}} + P_{\text{ely}} + P_{\text{sab}} + w_{\text{comp}}r_{\text{calc}}}_{\text{loads}} + \underbrace{P_{\text{ch}} - P_{\text{dis}}}_{\text{battery}} - \underbrace{(1 - \gamma)P_{\text{ac}}}_{\text{array}} = 0. \quad (27)$$

A regulatory layer below the controller enforces this in the real plant: it caps PV and battery to what the hardware can deliver and, when demand still exceeds supply, **sheds load in a fixed priority order** — calciner, contactor, electrolyser, Sabatier — recording every intervention and every undervoltage trip. A controller that needs the bus to save it is failing.

2 Linear approximation for the dispatch LP

The outer layer is a MILP over $n = 240$ hourly intervals: 7447 columns, 6964 rows, 24 integers. It carries *all* the economics and, apart from the kiln, no dynamics. See `docs/DISPATCH_NMPC.md §2` for the row census.

2.1 What is dropped, and what that costs

Full model	LP treatment	Consequence
16 states	7 states	fast thermal states invisible
$\eta_{\text{cap}}(V), \eta_{\text{act}}(i), X(T)$	concave PWL in dispatch	exact relaxation (§2.3)
reactor T , electrolyser T	dropped	light-off and warm-up unmodelled
kiln T	kept — balance is linear	5.6 h warm-up preserved
$r_{\text{calc}}(T)$ Arrhenius×Baker	linear bound + binary	genuinely non-convex (§2.4)
catalyst, stack degradation	dropped	no thermal-cycling cost
hourly grid	—	sub-hourly buffering invisible

The last line is not a rounding error. Measured closed-loop, the LP’s planned operating profit is flat across a 15× range of battery capacity while realised profit rises 2.5×: an hourly average of supply and demand simply does not contain the deficits a battery exists to cover, so the layer cannot price storage. See `experiments/BATTERY_SIZING.md`.

2.2 States and controls

$z_k \in \mathbb{R}^7 = (\sigma, n_{\text{CaCO}_3}, \bar{N}, n_{\text{H}_2}, n_{\text{CO}_2}, m_w, T_{\text{kiln}})$; 24 columns per interval: four commitments e_j , seven PWL segment fills, kiln power/rate/indicator, four start indicators, two battery powers, curtailment, two vent rates and make-up water.

2.3 Piecewise-linear rate maps

For the contactor, electrolyser and reactor, each nonlinear rate curve is replaced by segment fills $\delta_{j,m} \in [0, w_{j,m}]$:

$$d_j = \sum_m \delta_{j,m}, \quad r_j = \sum_m a_{j,m} \delta_{j,m}, \quad \delta_{j,m} \leq w_{j,m} e_j, \quad (28)$$

with slopes $a_{j,m}$ **strictly decreasing**. That concavity is what makes the relaxation *exact*: an optimiser maximising output fills the steepest segment first unprompted, so no ordering binaries are needed. The builder raises if a sampled curve is not concave — without it the formulation is silently wrong rather than merely approximate. Electrical draw is $P_j = \text{idle}_j e_j + d_j$ for the power-dispatched machines.

2.4 Kiln: the one irreducible binary

The energy balance is linear exactly as written in the full model, with the sensible-heat term evaluated at the operating temperature rather than at T_k — the only approximation, worth a few percent of duty during warm-up:

$$C \frac{T_{k+1} - T_k}{\Delta t} = \eta P_k^{\text{kiln}} - UA(T_k - T_a) - \Delta H r_k^{\text{cal}}. \quad (29)$$

The *rate* is the problem. The set $0 \leq r \leq \max(0, \alpha(T - T_{\text{on}}))$ is genuinely non-convex, and writing the cap directly rearranges to $T \geq T_{\text{on}} + r/\alpha$, which at $r = 0$ demands a permanently hot kiln. An indicator is unavoidable:

$$r \leq \bar{r} y, \quad r \leq \alpha(T - T_{\text{on}}) + M(1 - y), \quad y \leq e, \quad M = \alpha(T_{\text{on}} - 250). \quad (30)$$

Relaxing y to $[0, 1]$ permits $r^* = \bar{r}(\alpha(T - T_{\text{on}}) + M)/(\bar{r} + M)$, which at 841 K allows 0.26 mol/s where the real kiln calcines nothing — and reproduces exactly the failure it was added to prevent: calcination scheduled on a cold kiln, sorbent saturation, production collapsing to 17 kg/day. y is therefore binary over the first 24 hours and relaxed beyond, where only the terminal inventory depends on it.

2.5 Dynamics

Seven equalities per interval, all linear:

$$\sigma_{k+1} = \sigma_k \left(1 - \frac{\Delta t \kappa_{\text{sd}}}{86400}\right) + \frac{\Delta t \eta_c P_k^{\text{ch}}}{E} - \frac{\Delta t P_k^{\text{dis}}}{\eta_d E}, \quad n_{k+1}^{\text{CaCO}_3} = n_k^{\text{CaCO}_3} + \Delta t(r_k^{\text{cont}} - r_k^{\text{cal}}), \quad (31)$$

$$\bar{N}_{k+1} = \bar{N}_k + \Delta t r_k^{\text{cal}} / n_{\text{tot}}, \quad n_{k+1}^{\text{H}_2} = n_k^{\text{H}_2} + \Delta t(r_k^{\text{ely}} - 4r_k^{\text{sab}} - v_k^{\text{H}_2}), \quad (32)$$

$$n_{k+1}^{\text{CO}_2} = n_k^{\text{CO}_2} + \Delta t(r_k^{\text{cal}} - r_k^{\text{sab}} - v_k^{\text{CO}_2}), \quad m_{k+1}^w = m_k^w + \Delta t(\dot{m}_k^w + 2\phi_r M r_k^{\text{sab}} - M r_k^{\text{ely}}). \quad (33)$$

2.6 Remaining constraints

Bus balance (the block whose dual is λ), written demand-minus-supply so relaxing it means energy from nowhere and the dual is a positive price: $\sum_j P_{j,k} + w_{\text{comp}} r_k^{\text{cal}} + P_k^{\text{ch}} - P_k^{\text{dis}} = (1 - \gamma_k) \text{PV}_k$.

Reactor availability, a linear envelope of the plant's $\min(\cdot, \cdot)$ taper — needing no indicator, because the taper reaches zero exactly *at* the box floor so $r^{\text{sab}} = 0$ stays feasible:

$$r_k^{\text{sab}} \leq \bar{r}_{\text{sab}} \frac{n_k^{\text{CO}_2} - \underline{n}_{\text{CO}_2}}{50}, \quad r_k^{\text{sab}} \leq \bar{r}_{\text{sab}} \frac{n_k^{\text{H}_2} - \underline{n}_{\text{H}_2}}{200}. \quad (34)$$

Commitment: minimum electrolyser load $\sum_m \delta_{\text{ely},m,k} \geq \underline{d} e_k$; start counting $s_{j,k} \geq e_{j,k} - e_{j,k-1}$, with interval 0 measured against what is *already running* rather than against zero; and minimum up-time $\tau_j s_{j,k} \leq \sum_{l=k}^{k+\tau_j-1} e_{j,l}$ with $\tau = 3$ for the calciner and 2 for the reactor.

2.7 Objective

$$\min \sum_k \left[-p_{\text{CH}_4} M_{\text{CH}_4} \Delta t r_k^{\text{sab}} + \frac{c_{\text{sorb}} \Delta t r_k^{\text{cal}}}{n_{\text{tot}}} + \frac{c_{\text{batt}} \Delta t}{2E} (P_k^{\text{ch}} + P_k^{\text{dis}}) + c_w \Delta t \dot{m}_k^w + \sum_j c_j^{\text{start}} s_{j,k} \right] - V(z_n), \quad (35)$$

with start costs 0.50/40/5/40 EUR for contactor/calciner/electrolyser/reactor and terminal value V pricing leftover inventory at the methane it would become. Without V the layer empties every buffer at the horizon end, which is locally optimal and globally wrong.

$c_{\text{batt}} = \text{CAPEX} / (E \cdot \text{DoD}_{\text{ref}} \cdot N_{\text{cyc}} \cdot \eta_{\text{rt}}) = 0.0565 \text{ EUR/kWh}$ at the reference pack, against a night-time shadow price of $\approx 0.043 \text{ EUR/kWh}$. p_{CH_4} **is a control tuning parameter, not a market price**: set below the crossover the optimiser declines to cycle the battery, curtails a third of the array, runs the pack flat and trips the bus. It is fixed at 5.50 EUR/kg, above the crossover with margin.